

Sixth Information Systems International Conference (ISICO 2021)

Business process analysis of programmer job role in software development using process mining

Rokhman Fauzi*, Rachmadita Andreswari

Telkom University, Jl. Telekomunikasi no.1, Bandung, 40257, Indonesia

Abstract

In software development, the contribution of each role is needed to run the business process according to the expected goals. This research was conducted to identify the programmer's contribution in the software engineering course. Based on the record of commit data through collaboration tools, it was found that programmers contributed less when compared to project managers and analysts. So, it is necessary to discover the programmer's workflow. Process mining techniques were used in this study to obtain a flow based on the event log generated by the collaboration tools. The results of this study were on average; each team carried out 4-6 activities. Based on SEQUAL framework, whether syntactic, semantic, or pragmatic, there are issues that occur, including deadlocks, lack of synchronization, superfluous activity, labeling issues, line crossing, and crooked alignment. Handling the quality issue of the process can improve the role of programmers and all members of the software development team.

© 2021 The Authors. Published by Elsevier B.V.

This is an open access article under the CC BY-NC-ND license (<https://creativecommons.org/licenses/by-nc-nd/4.0>)

Peer-review under responsibility of the scientific committee of the Sixth Information Systems International Conference.

Keywords: process mining; business process; software development; job role; programmer

1. Introduction

The success or failure of a software development project is significantly influenced by human resource factors [1]. Therefore, the competence of the development team members is one of the most critical factors for success. In software development, competencies consist of two parts, namely hard competencies and soft competencies [2]. Hard competence is competency related to technical skills in software development. Meanwhile, soft competencies describe competencies related to individuals' behavior, creativity, and innovation [2]. These competencies are needed in optimizing project workflows according to the job roles of each team member.

* Corresponding author.

E-mail address: rokhmanfauzi@telkomuniversity.ac.id

In software development projects in the Software Engineering course, there are seven development teams. Each development team consists of three job roles [3], namely project manager, analyst, and programmer. The programmer has a large enough workload so that there are three programmers in each team. We obtained preliminary information that some programmers are not working according to their job roles through our observations. Through this research we investigate programmer's workflow based on the event log generated by the collaboration tools.

This study aims to find the weaknesses of the programmers workflow in seven development teams. The analysis is carried out on the programmer's workflow in the collaboration tools which is represented by the process models. We used process mining techniques to discover this issue. It is expected that a process model of seven development teams can be obtained from a process discovery, then the workflow of each team can be analyzed. First, we generate process models using Apromore and then evaluate their quality [4] in semantic, syntactic, and pragmatic issues using the SEQUAL framework [5] [6]. Each activity executed by an actor on the development team affects the model process. The event log influences the BPMN model produced by Apromore. Furthermore, it is expected that the programmer's workflow can be analyzed using quality analysis in the SEQUAL framework. Therefore, the quality assurance of each development team's workflow can be identified. The evaluation of the quality of this business process model will be one of the suggestions for improving the Software Engineering learning process in the coming year.

2. Related works

Process mining is a technique for discovering, monitoring, and improving the actual process based on the event log contained in an information system. Using process mining techniques, the researcher can obtain knowledge from the extracted logs [7]. Several studies have been conducted to analyze software development business processes using process mining. Process mining is implemented in several business processes in example healthcare, enterprise resource planning, education, and other business processes recorded in an information system. In the field of software development, the development team's process discovery has been carried out in the study [8] and [9].

In a study [9], software development process discovery was carried out to measure the efficiency of team performance. The existence of this research is expected to also know the potential hidden costs, risks, and time so that it allows for improvement. This study provides analysis from a control-flow point of view to analyze the process model obtained from tracking software development on a simulated team (at the level 3 lecture at the undergraduate stage). The development of this research can be done by comparing the process carried out with the resulting artifacts and predicting good and bad models.

Business Process Modeling Notation (BPMN) quality assurance measurement was assessed in the previous research by several researchers. Based on [10] study, the quality of the BPMN model was checked to get feedback, whether the model made has good or bad quality. The first stage is the creation of a process model using BPMN. Furthermore, an evaluation using quality measurements, including quality characteristics, quality metrics, quality criteria, and quality functions. In the final section, the evaluation results of the process model that have been assessed for quality are explained. This research develops a quality evaluation model of the process modeling using BPMN, namely MAQ. The level of confidence of the model made was 95% based on the student's t-test. MAQ only measures the good or bad quality of a process model and produces absolute numbers from the results of its assessment.

Furthermore, this study [6] aims to find out how humans check the quality of BPMN in a case study. This study compares the checks of two different process models, which then checks the quality of the process models using the SEQUAL framework [17], namely syntactic, semantic and pragmatic. The survey conducted in this study involved several people with different levels of understanding and track records based on defined demographics. This study indicates that humans can adapt strategies in identifying the quality of the process model. Many of these quality problems are identified by the subject, in this case, namely humans, but some do not get enough attention.

Moreover, research on the construction of business process models has produced various techniques to assess the quality of these models. The general description and consensus are not yet sufficiently available on the various techniques for assessing the quality of the business process model. Furthermore, [11] study is conducted to collect, analyze, organize, and integrate existing knowledge in a comprehensive framework that seeks to balance completeness and relevance without obstructing the general picture. This research resulted in the Comprehensive Process Model Quality Framework (CPMQF). This framework summarizes 39 quality dimensions, 21 quality metrics, 28 quality

(sub) drivers, 44 (sub) driver metrics, 64 realization initiatives, and 15 concrete process model objectives related to the four types of organizational benefits and the relationships between these. This review is thus considered a valuable instrument for researchers and practitioners concerned with the quality of process models.

Based on those previous studies, this study applies process mining techniques in education to find out the flow carried out by programmers in software development courses. Furthermore, from the result of the process model, SEQUAL framework is implemented to assess the quality of the process model produced by programmer activities. Humans can carry out an assessment of the quality of this process model based on issue quality in research [6]. According to research, the purpose of assessing the quality of this process model is to find weaknesses in the process carried out by programmers, according to research [11].

3. Related works

This research begins with event-log cleansing. We identify the workflow team in software development. By the objectives of this study, we also identify programmer's workflow. Then we discover the process model of the programmer using the process mining technique with Apromore tool. The process model is generated in BPMN.

In the next step, we perform quality assurance using the SEQUAL framework covering syntactic, semantic, and pragmatic. The description of quality assurance refers to [5] and [4]. Finally, we discover and conduct process weakness model analysis. The following is the flow of this research (Fig. 1)

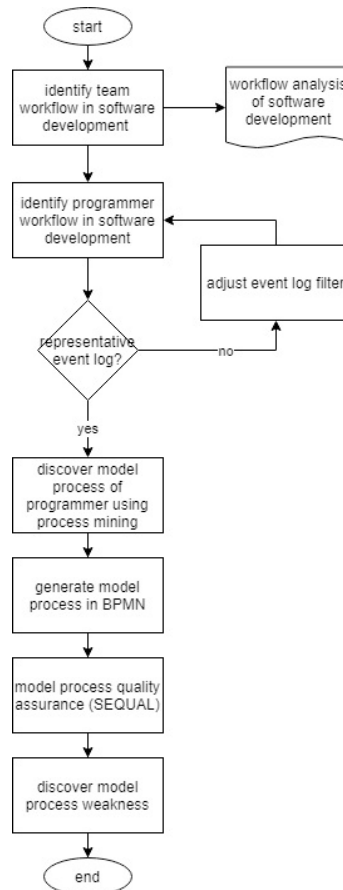


Fig. 1. Systematic research.

Based on Fig. 1, this study is conducted based on the previous data processing. The analysis was carried out for all software development processes. While in this study, only handling the programmer's flow. In the previous processing stage, programmers were deemed less contributing when viewed from the number of commits on a collaboration tool.

4. Result and discussion

In general, activities of each team are; Open, Implement Tasks, Resolved, Merge Branches and Close. The generation of these 5 activities was developed through preprocessing of the development team issues-solving in the Bitbucket tools. In each team, there are three job roles, namely project manager, programmer, and analyst. Then, the event log is filtered by the job role programmer, and the activities remaining are; Open, Implement Task, Resolved, and Merge Branch. Ideally, every programming activity in the Capstone project course is executed by programmers, yet the software development process is often done by another job role. The following is the example of programmer contribution from the two software development teams (Table 1). Percentage of each actor obtained from the Apromore tools by comparing total cases done by the actor and overall cases.

Table 1. Contribution of each job role to a team.

No	Job Role	Team	Sum of Case	Percentage
1	ANA1	team 1 (minjemin)	22	75.86%
2	ANA2	team 1 (minjemin)	5	17.24%
3	PGM1	team 1 (minjemin)	7	24.14%
4	PGM2	team 1 (minjemin)	7	24.14%
5	PGM3	team 1 (minjemin)	6	20.69%
6	PM	team 1 (minjemin)	15	51.72%
7	ANA1	team 2 (titipmasa)	8	29.63%
8	ANA2	team 2 (titipmasa)	5	18.52%
9	PGM1	team 2 (titipmasa)	5	18.52%
10	PGM2	team 2 (titipmasa)	5	18.52%
11	PGM3	team 2 (titipmasa)	6	22.22%
12	PM	team 2 (titipmasa)	11	40.74%

Table 2. Software development team event log (Team 1).

Case ID	Activity	Resource	Job Role	Timestamp
1	Start	Rizky Rahmat Hakiki	PM	19/02/2019 18:38
1	Merge Branch	danti puji yuherisna	PGM3	24/02/2019 18:00
1	Implement task	danti puji yuherisna	PGM3	24/02/2019 18:00
1	Implement task	danti puji yuherisna	PGM3	27/02/2019 18:00
1	End	Alif Jafar	ANA1	31/03/2019 19:00
2	Start	Alif Jafar	ANA1	18/02/2019 18:00
2	Merge Branch	danti puji yuherisna	PGM3	31/03/2019 19:00
2	Merge Branch	danti puji yuherisna	PGM3	31/03/2019 19:00
2	End	danti puji yuherisna	PGM3	31/03/2019 19:00
3	Start	Ryan Risbaya	PGM2	24/02/2019 18:00
3	Implement task	Ryan Risbaya	PGM2	24/02/2019 18:00
3	Implement task	firdha utami	PGM1	24/02/2019 18:00
3	Open	firdha utami	PGM1	24/02/2019 23:30
3	Resolved	firdha utami	PGM1	24/02/2019 23:34
3	End	firdha utami	PGM1	24/02/2019 23:34

Based on Table 1, it can be seen that in each software development team, programmers do not have a dominant role, especially in contributing to the number of activities carried out on system development. Therefore, the role of a programmer in a software development collaboration should contribute more to the resolution of an issue

through collaboration tools. Moreover, the following is an example of the event log generated in a software engineering course. In this case, the activity conducted by all job roles is selected (Table 2). There are seven event logs from each development team that are modeled using Apromore tools. Process model generated in the form of BPMN in order to analyze the quality of the business process.

The event log is then processed using Apromore to get the process carried out by each team. The process of getting an existing flow is being executed using the first process mining technique called process discovery. Using the tools, resources of the software development team are filtered by only programmer role. Based on the process discovery, the programmer's workflow in software development can be discovered and analyzed. This process discovery produces a process model in BPMN with "average" duration and "relative-case" frequency settings in Apromore. Otherwise, the arc or edge is set to a full scale of 100%. The following is the BPMN process model of the process discovery for all teams (Fig. 2 – Fig. 8)

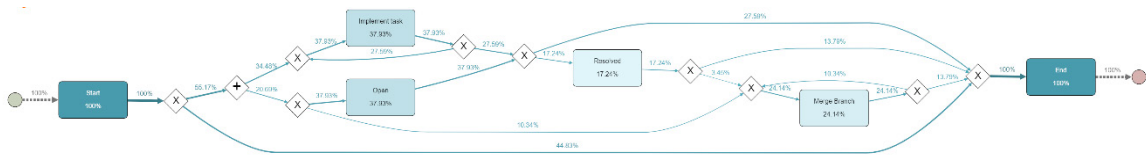


Fig. 2. Model Process of Team 1 (Minjemini).

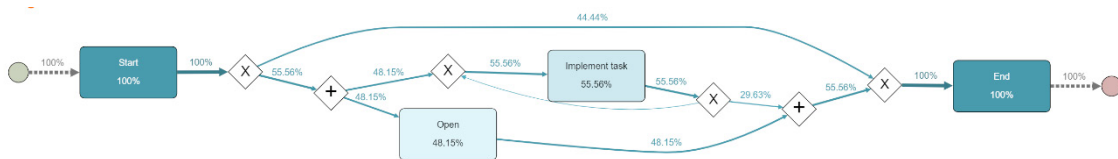


Fig. 3. Model Process of Team 2 (Titipmasa).

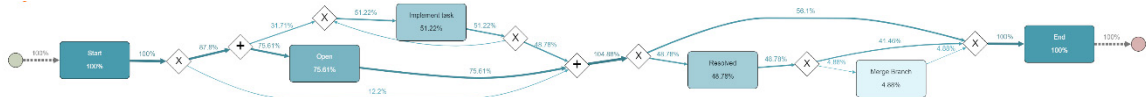


Fig. 4. Model Process of Team 3 (Xplore).

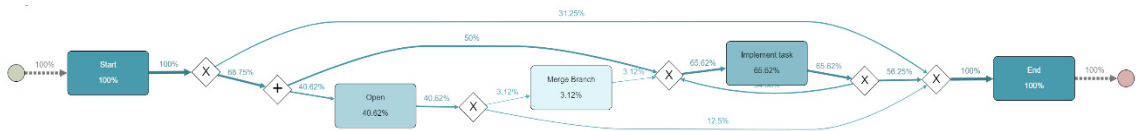


Fig. 5. Model Process of Team 4 (Sportloka).

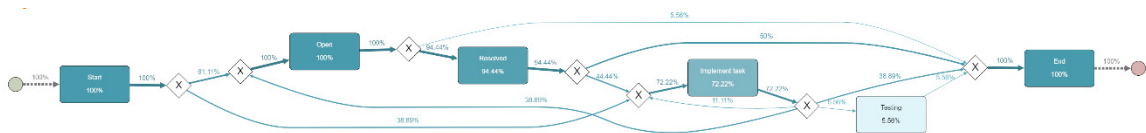


Fig. 6. Model Process of Team 5 (Farming).

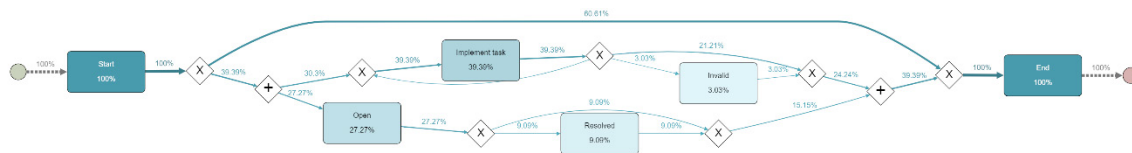


Fig. 7. Model Process of Team 6 (Suapin).

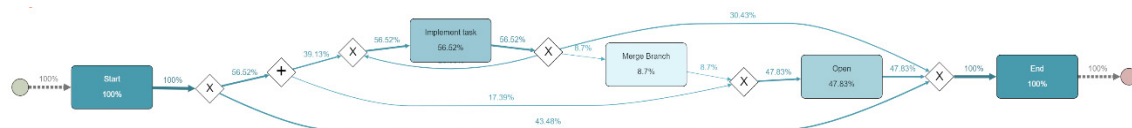


Fig. 8. Model Process of Team 7 (Doctor).

Based on the BPMN modeling results, characteristics analysis of each process model is carried out. The analysis was done based on the number of activities, gateways, and edges in each BPMN process model. In each of the resulting process models, the number of activities carried out by programmers is different, depending on their performance in the team; for example, in team 1, team 3, team 5, and team 6, there are six activities. This number of activities is the first thing checked on the process model because it shows the involvement of activities carried out by actors in an activity according to their job desk. Moreover, the complexity analysis of the process model can be seen at the gateway and edge. There are generally two gateways produced, namely exclusive gateways and parallel gateways. The exclusive gateway shows the selection of one activity, while the parallel gateway shows two activities that are executed together. In Fig. 2 - Fig. 8, the more gateways and edges, the more complicated the process model. Overall, the result of BPMN characteristics can be seen in Table 3.

Table 3. Characteristics of the development team process model.

Team	Sum of activity	Sum of gateway	Sum of edge
team 1 (minjemini)	6	10	21
team 2 (titipmasa)	4	6	11
team 3 (xplore)	6	8	19
team 4 (sportloka)	5	6	14
team 5 (farming)	6	7	19
team 6 (suapin)	6	9	19
team 7 (doctor)	5	6	14

Table 3 shows that the number of activities performed by programmers is in the range of 4-6 activities, with gateways of 6-10 and edges of 11-21. In general, the number of gateways affects the edges of each process model, thus affecting the complexity of the BPMN generated. Furthermore, analyses were carried out related to the SEQUAL framework, namely syntactic, semantic and pragmatic, based on research conducted by [6]. Identification of syntactic, semantic and pragmatic issues is based on descriptions of related references, including [4], [6], [12], [13], [14], [15], and [16]. Process model identification is conducted by concerning the quality issue. The results of the identification of issues on quality assurance as a whole can be seen in Table 4.

Based on the identification of syntactic issues, three issues are obtained from the whole team, namely the lack of synchronization. These problems were identified in Team 1, Team 4, and Team 7. In addition, the issue of deadlock occurred in Team 3. Furthermore, the semantic superfluous activity problem was in Team 5 and Team 6. In pragmatic problems, there were three issues occurring in the three different teams, namely label issues (Team 6), line crossing (Team 5), and crooked alignment (Team 5).

Table 4. The number of issues business process quality assurance.

QA Issues	Team 1 (minjemini)	Team 2 (titipmasa)	Team 3 (xplore)	Team 4 (sportloka)	Team 5 (farming)	Team 6 (suapin)	Team 7 (doctor)
Syntactic issues							
Wrong modeling element usage	0	0	0	0	0	0	0
Missing transition conditions	0	0	0	0	0	0	0
Deadlock	0	0	1	0	0	0	0
Livelock	0	0	0	0	0	0	0
Lack of synchronization	1	0	0	1	0	0	1
Semantic issues							
Superfluous activity	0	0	0	0	1	1	0
Invalid behavior	0	0	0	0	0	0	0
Switched lane labels	0	0	0	0	0	0	0
Pragmatic issues							
Label issues	0	0	0	0	0	1	0
Line crossings	0	0	0	0	1	0	0
Message flow descriptions	0	0	0	0	0	0	0
Compact layout	0	0	0	0	0	0	0
Erratic sequence flow direction	0	0	0	0	0	0	0
Reverse sequence flow direction	0	0	0	0	0	0	0
Implicit gateways	0	0	0	0	0	0	0
Crooked alignment	0	0	0	0	1	0	0
All issues	1	0	1	1	3	2	1

Based on the assessment of the quality of the process model, it was found that Team 5 had the most quality issues, namely superfluous activity, line crossing, and crooked alignment. In addition, the quality assurance assessment for Team 6 shows two issues, namely: superfluous activity and label issues. All in all, the quality assurance issues of each process model are affected by the collaboration process of the software development team in solving issues. Moreover, the algorithm of the Apromore also has a contribution in the process model that is formed.

5. Conclusion

The discovery process carried out in this study was conducted to determine the flow carried out by programmers on the software development team. The results obtained through the Apromore application are a process model in BPMN notation. Overall, all programmers carry out activities, namely Open, Implement Task, Resolved, Merge Branch, Invalid, and Testing. However, not all of these activities were carried out on each team. On average, each team carried out 4-6 activities (including start and end). It shows that programmers have not played a comprehensive role in software development, especially the coding process using collaboration tools. This is because each of these activities represents an issue-solving activity in software development, where the programmer is the main role. In addition, in this capstone project course, the number of commits by the development team has a significant component of the assessment. In the future, it is necessary to revise the assignment of job desk programmers so that they can contribute according to their job role.

Furthermore, the assessment results based on the SEQUAL (semiotic quality) framework show that the programmer's business processes have several issues related to quality, both syntactic, semantic, and pragmatic. Based on the identification of this issue, it was found that the weakness of the process carried out when viewed from the quality aspect of the process model, that is more in the syntactic aspect. Two issues occur in the four teams, namely,

the issue of deadlock and lack of synchronization. In addition, there are issues of superfluous activity in the semantic aspect, and in the pragmatic aspect, there are issues of labelling issues, line crossing, and crooked alignment. Handling the quality issue of the process can be used to improve the role of programmers and all members of the software development team.

References

- [1] Discenza, R., and J. B. Forman. (2007) "Seven causes of project failure: how to recognize them and how to initiate project recovery," PMI® Global Congress 2007, Project Management Institute.
- [2] Holtkamp, Philipp. (2015) "Competency Requirements of Global Software Development Conceptualization, Contextualization, and Consequences,," *Journal of Universal Science*, 2015.
- [3] T. Kusumasari, "Competency Profile for Software Development Team that Support Project Success," *International Journal on Advanced Science, Engineering and Information Technology* **10**.
- [4] Dumas, Marlon, M. La Rosa, J. Mendling and H. A. Reijers. (2018) "Fundamentals of business process management" (2nd ed.), Berlin, Springer.
- [5] Krogstie, John. (2021) "Quality of Business Process Models" *Lecture Notes in Business Information Processing* **134**.
- [6] Haisjackl, C, P. Soffer and L. S. Y., "How do humans inspect BPMN models: an exploratory study," *Software System Model* **17 (2)**: 655–673.
- [7] Aalst, W. V. D. (2016) "Process Mining: Data Science in Action", Berlin, Springer
- [8] Marques, Rita, M. Mira da Silva and D. Ferreira. (2018) "Assessing Agile Software Development Processes with Process Mining: A Case Study," *IEEE 20th Conference on Business Informatics (CBI)*.
- [9] Caldeira, João, F. Brito e Abreu, J. Reis and J. Cardoso. (2019) "Assessing Software Development Teams' Efficiency using Process Mining," in *International Conference on Process Mining (ICPM)*, Aachen.
- [10] Sadowska, Małgorzata. (2013) "Quality of business models expressed in BPMN".
- [11] Meyer, P. De and C. J. (2018) "An overview of process model quality literature - The Comprehensive Process Model Quality Framework," *arXiv preprint arXiv:1808.07930*.
- [12] Mendling, J, H. Reijers and J. Recker. (2010) "Activity labeling in process modeling: empirical insights and recommendations,," *Information System* **35(4)**: 467–482.
- [13] Purchase, Helen. (1997) "Which aesthetic has the greatest effect on human understanding?," in *Proceedings of GD'97*, Berlin, Springer.
- [14] Gschwind, Thomas, J. Pinggera, S. Zugall, H. A. Reijers and B. Weber. (2012) "A Linear Time Layout Algorithm for Business Process Models. Technical Report," IBM Research.
- [15] Weilkiens, T, C. Weiss, A. Grass and K. N. Duggen. (2016) "Modeling Business Processes Using BPMN, Morgan Kaufmann".
- [16] Schrepfer, Matthias, J. Wolf, J. Mendling and H. A. Reijers. (2016) "The impact of secondary notation on process model understanding," *Proceedings of 2nd Working Conference on The Practice of Enterprise Modeling*.
- [17] Lindland, Odd Ivar, G. Sindre and A. Solvberg. (1994) "Understanding Quality in Conceptual Modeling," *IEEE Software* **11 (2)**: 42–49.